DO YOU KNOW...?

Vitamins are the discovery of 20th century scientists. In the history of nutrition, the findings of Casmir Funk, 1911 was a turning point who propounded the 'Vitamine' theory by feeding animals purified diet. Takaki, a physician in the Japanese Navy, first demonstrated that fatal diseases like beriberi could be treated with rice bran, vegetables, fish and meat. Thus the study of vitamins brought to light about 17 different vitamins. Each one of them has its own history, chemistry, structure, function, sources, requirements and disorder symptoms.

- know the important sources and deficiency symptoms of vitamins, minerals and water.
- importance of water in day to day life.

11.1 Classification of Vitamins

Vitamins differ from each other in physiological function, chemical structure, and distribution in foods. Plants synthesize all vitamins they require and therefore vegetables and fruits are rich sources of vitamins. They are broadly divided into two categories, on the basis of solubility in fats or water.

11.2 Fat Soluble Vitamins

Fat – soluble vitamins are soluble in fats and fat solvents. They are insoluble in water. So these vitamins are utilized only if there is enough fat in the body.

11.2.1 Vitamin A

Vitamin A was discovered in 1909 and its chemical name is retinol. Vitamin A compounds include retinol, retinal and retinoic acid. It has a specific function in the retina of the eye. Vitamin A occurs only in foods of animal origin. Vitamin A activity is possessed by carotenoids found in plants. Hence carotenoids are called Provitamin A.

Provitamins are substances that are chemically related to a vitamin but it must be changed by the body into the active form of the vitamin. Carotene is known as precursor of Vitamin A.

Functions

- It provides the required stimulation for vision in the retina and is essential for maintaining normal vision.

Table 11.1 Classification of Vitamins

Fat soluble vitamins	A, D, E and K. Fat soluble vitamins are stored in the liver and fatty tissues. These are not readily excreted from the body.
Water soluble vitamins	B (B_1, B_2, B_3, B_5, B_6, B_{12}) and C Water soluble vitamins travel in the blood and are stored in limited amounts. These are readily excreted from the body through urine.

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